Name: _____ Register No: _____

Experiment 8

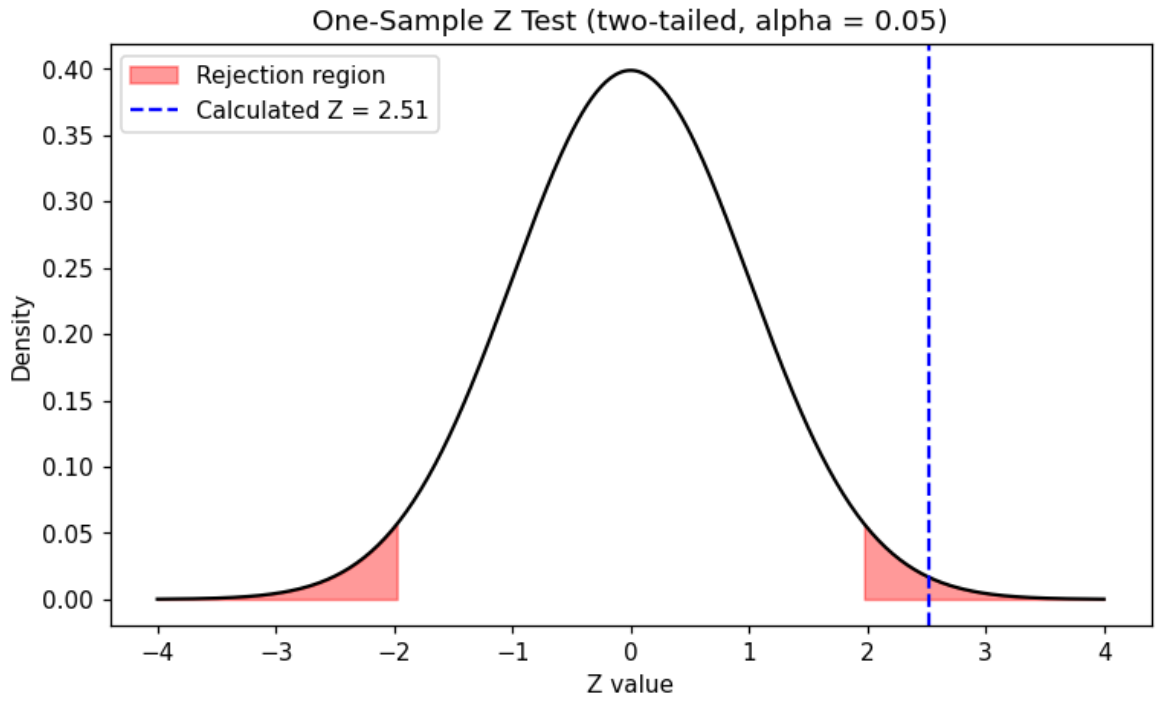
Question: Conduct Z test for the data given using Python.

Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats

def exp_8():
    from statsmodels.stats.weightstats import ztest
    rng = np.random.default_rng(5)
    alpha = 0.05

    # ---------- One-sample Z test ----------
    # Claim: average battery life is 500 hours. Known population sigma = 40.
    mu0, sigma = 500, 40
    sample = rng.normal(loc=520, scale=sigma, size=64).round(1)
    n, xbar = len(sample), sample.mean()
    z = (xbar - mu0) / (sigma / np.sqrt(n))
    p = 2 * (1 - stats.norm.cdf(abs(z)))
    z_crit = stats.norm.ppf(1 - alpha / 2)

    print("ONE-SAMPLE Z TEST")
    print("H0: mu = 500        H1: mu != 500   (two-tailed, alpha = 0.05)")
    print(f"n = {n}, sample mean = {xbar:.2f}, sigma = {sigma}")
    print(f"Z statistic      = {z:.4f}")
    print(f"Critical value   = +/-{z_crit:.3f}")
    print(f"p-value          = {p:.4f}")
    print("Decision         :", "Reject H0" if p < alpha else "Fail to reject H0")
    zs, ps = ztest(sample, value=mu0)
    print(f"statsmodels check (uses sample std): z = {zs:.4f}, p = {ps:.4f}\n")

    # ---------- Two-sample Z test ----------
    # Compare average test scores of two classes.
    class_a = rng.normal(72, 10, 80).round(1)
    class_b = rng.normal(68, 10, 90).round(1)
    se = np.sqrt(10**2 / len(class_a) + 10**2 / len(class_b))   # known sigma = 10
    z2 = (class_a.mean() - class_b.mean()) / se
    p2 = 2 * (1 - stats.norm.cdf(abs(z2)))
    print("TWO-SAMPLE Z TEST")
    print("H0: mu_A = mu_B     H1: mu_A != mu_B   (two-tailed, alpha = 0.05)")
    print(f"Class A: n={len(class_a)}, mean={class_a.mean():.2f}")
    print(f"Class B: n={len(class_b)}, mean={class_b.mean():.2f}")
    print(f"Z statistic      = {z2:.4f}")
    print(f"p-value          = {p2:.4f}")
    print("Decision         :", "Reject H0" if p2 < alpha else "Fail to reject H0")
    zs2, ps2 = ztest(class_a, class_b)
    print(f"statsmodels check (uses sample std): z = {zs2:.4f}, p = {ps2:.4f}")

    # ---------- Plot rejection region ----------
    x = np.linspace(-4, 4, 400)
    y = stats.norm.pdf(x)
    plt.figure(figsize=(8, 4.5))
    plt.plot(x, y, color="black")
    plt.fill_between(x, y, where=(x <= -z_crit), color="red", alpha=0.4, label="Rejection region")
    plt.fill_between(x, y, where=(x >= z_crit), color="red", alpha=0.4)
    plt.axvline(z, color="blue", linestyle="--", label=f"Calculated Z = {z:.2f}")
    plt.title("One-Sample Z Test (two-tailed, alpha = 0.05)")
    plt.xlabel("Z value")
    plt.ylabel("Density")
    plt.legend()
    plt.show()


exp_8()
```

Output

```
ONE-SAMPLE Z TEST
H0: mu = 500        H1: mu != 500   (two-tailed, alpha = 0.05)
n = 64, sample mean = 512.57, sigma = 40
```

```
Z statistic      = 2.5144
Critical value   = +/-1.960
p-value          = 0.0119
Decision         : Reject H0
statsmodels check (uses sample std): z = 2.6726, p = 0.0075

TWO-SAMPLE Z TEST
H0: mu_A = mu_B     H1: mu_A != mu_B   (two-tailed, alpha = 0.05)
Class A: n=80, mean=71.53
Class B: n=90, mean=68.65
Z statistic      = 1.8746
p-value          = 0.0608
Decision         : Fail to reject H0
statsmodels check (uses sample std): z = 1.8478, p = 0.0646
```

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